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Correlation Alignment in Filter Bank Riemannian Tangent Space for Motor Imagery Classification

Xiaokun Zou[1]
College of Automation Science and Engineering
South China University of Technology
Guangzhou, China
1833902777@qq.com

Xiaofeng Xie[2]
Mechanical And Electrical Engineering College
Hainan University
Haikou, China
xfxie@hainanu.edu.cn

Feifei Qi*
School of Internet Finance and Information Engineering
Guangdong University of Finance
Guangzhou, China
20-081@gduf.edu.cn

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***Abstract*—In motor imagery brain-computer interface (BCI), the spatial covariance matrices of electroencephalography (EEG) signals which is lying on smooth Riemannian manifolds, are well used to enhance the performance of motor imagery classification. However, in different session, due to the changes of spatial and frequency information which caused by physiological, environmental, as well as instrumental changes, model trained in calibration session often get poor performance in use session. To improve the cross-session performance, combining unsupervised domain adaptive methods with Riemannian manifold, we propose a method called as Correlation Alignment in Filter Bank Riemannian Tangent Space (FBTSCORAL). First, raw EEG signals are constructed into multiple frequency-bands spatial covariance matrices by filter bank, in each bands the spatial covariance matrices are mapped into Riemannian tangent space as a tangent vector, the final feature vector is merged by tangent vectors in all bands. Second, Correlation Alignment (CORAL) is used to align feature space in calibration session and feature space in use session. Finally, SVM is used to fit the aligned feature vectors. The experiment shows that our work achieves the state-of-the-art performance in BCI competition IV 2a dataset and the second performance in BCI competition III 3a and 4a datasets.**

Keywords- motor imagery; Riemannian manifold; feature by Riemannian tangent space; Correlation Alignment

I. INTRODUCTION

BCI is a system which can identify cognitive states and intent by decoding neurophysiological signals [1]. It can be used to control a wheelchair, manipulator or mouse cursor by decoding the EEG signal. Our work focuses on motor imagery systems which obtain EEG signals by imagining movements of body (e.g., the right hand, left hand, foot and tongue) [3]. The motor imagery EEG signals which include richly spatial and spectral information can help to recognize the movement intent of body.

In the literature many methods have been proposed to extract spatial and spectral features from motor imagery EEG signals. The CSP-based methods extract features by spatial filter which maximize variance in one condition while minimizing variance in another condition [6][7]. Classic CSP mothed got the spatial filter from a large bandpass frequency (such as 8-35 Hz), which includes α and β waves related to motor imagery [8]. However, a large bandpass frequency band cannot discriminate the contribution of α or β wave. Several develop of CSP were proposed to tackle the problem. FBCSP used multiple frequency bands and got spatial filter on each band respectively [9]. SBCSP used a Gabor filter on multiple bands [10]. AFBCSP adaptively selected subject-specific discriminative bands [11]. Most CSP-based methods calculated center point by arithmetic mean. However, spatial covariance matrices lie in a non-Euclidean space naturally. The space of covariance matrices which is symmetric positive definite is a Riemannian manifold. Riemannian manifold-based methods construct spatial covariance matrices from EEG signals, where the relationship of samples is expressed by the Riemannian distance [2][12][13][14]. Many efficient tools of Riemannian geometry can be used to improve classification performance, such as Riemannian mean and Riemannian tangent space [2]. MDRM calculated the Riemannian mean points of each class in calibration session and assign the sample to the closest class by Riemannian distance in use session [2]. TSLDA extracted features by mapping the covariance matrices into tangent space, where the distance structure was consistent with the Riemannian manifold and the relationship between points was linear [2]. BSML designed a bilinear mapping method for dimensionality reduction in covariance matrices [13].

Although Riemannian manifold-based methods can obtain spatial features from EEG signals efficiently, they are mainly designed to project EEG signals into Riemannian manifold corresponding to a large bandpass frequency and without considering frequency band selection in Riemannian manifold mapping. In addition, the methods mentioned above did not consider the changes between calibration session and use session caused by physiological, environmental, instrumental changes and so on. In this paper, combining Riemannian manifold tools and unsupervised domain adaptation method Correlation Alignment [15], as well as filter banks, we propose **FBTSCORAL**. Our mothed achieves the state-of-the-art performance in BCI competition IV 2a dataset [18] and the second performance in BCI competition III 3a and 4a datasets [21].

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II. RELATED WORK

A. Construct spatial covariance matrices in BCI

motor imagery consists in the imagination of a limb movement like hand or foot. Different body parts are represented in different area of the motor cortex (roughly, right hand under C3 electrode, left hand under C4, foot under Cz, etc.) [2][16]. The EEG recording is usually cut into trials by short-time windows. We consider short-time trials $X_i = [x_1, \dots x_{T_s}] \in \mathbb{R}^{n \times T_s}$ which is the *i-th* trial or sample, where *n* denotes recording channels and T_s denotes time points in each trial [2]. For *i-th* trial, when $T_s \gg n$, an unbiased estimator of spatial covariance matrix is estimated using the Sample Covariance Matrix (SCM) $R_i \in \mathbb{R}^{n \times n}$ [2][17]:

$$R_i = \frac{1}{T_s - 1} X_i X_i^T, \tag{1}$$

B. Elements of Riemannian Geometry

The SCMs $R_i \in \mathbb{R}^{n \times n}$ lies in a space of symmetric positive-definite matrices (SPD). The SCMs is a differentiable Riemannian manifold M [2][13]. The concepts of Riemannian distance and tangent space play an important role in the application of Riemannian manifolds [2]. The Riemannian distance between R_1 and R_2 is defined as:

$$\delta_R(R_1, R_2) = \|\log R_1^{-1} R_2\|_F = \left[\sum_{i=1}^{n} log^2 \beta_i\right]^{\frac{1}{2}}, \tag{2}$$

Where R_1 and R_2 are SPD matrix, $\|*\|$ is the Frobenius norm and β_i is eigenvalue of $R_1^{-1} R_2$. Riemannian distance is the minimum length curve connecting two points on a Riemannian manifold.

Using Riemannian geodesic distance (2), the Riemannian mean point R is given by:

$$\mathfrak{G}(R_1, \dots, R_I) = \arg \min_{P \in P(n)} \sum_{i=1}^{I} \delta_R^2(R, R_i), \tag{3}$$

For a manifold of nonpositive sectional curvature like *P(n)*, such local minimum exists and is unique [19]. However, there is no closed-form expression to compute the mean. An efficient iterative method to compute the Riemannian mean of I SPD matrices is given in [20]. The tangent space of a Riemannian manifold is a linear space, which can often be used to study the nonlinearity of manifolds [2][12]. The tangent space $T(n)$ at point $\boldsymbol{R}$ is defined as:

$$T(n) = \left\{\boldsymbol{s_i} = upper\left(\boldsymbol{R}^{-\frac{1}{2}} Log_P(\boldsymbol{R_i}) \boldsymbol{R}^{-\frac{1}{2}}\right) \in \mathbb{R}^{\frac{n(n+1)}{2}}\right\}, \tag{4}$$

where $\boldsymbol{R}$ is a tangent point (Riemannian mean point of all samples in our work) and the *upper(*)* denotes the upper triangular part of matrix and vectorizes it. The log mapping is denoted by $Log_P(\boldsymbol{R_i}) = \boldsymbol{R}^{1/2} \log(\boldsymbol{R}^{-1/2} \boldsymbol{R_i} \boldsymbol{R}^{-1/2}) \boldsymbol{R}^{1/2}$. The Riemannian distance between $\boldsymbol{R}$ and nearby point $\boldsymbol{R_i}$ is about equal to the Euclidean distance on tangent space [2],

$$\delta_R(\boldsymbol{R}, \boldsymbol{R_i}) \approx \|\boldsymbol{s} - \boldsymbol{s_i}\|_{F.} \tag{5}$$

Generally, all samples from the dataset can be considered in a neighbor, while the mean of all samples is regarded as the tangent point $\boldsymbol{R}$ which calculated by (3).

C. Correlation Alignment

Correlation Alignment [15] is a simple but effective method for unsupervised domain adaptation, which minimizes domain shift by aligning the second-order statistics of source and target distributions. Labeled source-domain samples (calibration-session samples) $D_S = \{\boldsymbol{x_i}\}$, $\boldsymbol{x_i} \in \mathbb{R}^d$ with labels $L_S = \{y_i\}$, $y_i \in \{1, \dots, L\}$, and unlabeled target-domain samples (use-session samples) $D_T = \{\boldsymbol{u_i}\}, \boldsymbol{u_i} \in \mathbb{R}^d$ are given. Suppose all features are normalized to zero mean and unit variance, $\boldsymbol{u_S} = \boldsymbol{u_T} = 0$ but $\boldsymbol{C_S} \neq \boldsymbol{C_T}$. Here $\boldsymbol{u_S}, \boldsymbol{u_T}$ are feature vector mean and $\boldsymbol{C_S}, \boldsymbol{C_T}$ are covariance matrices. To minimize the distance between the second-order statistics (covariance matrices) of the source and target features, a linear transformation A is applied to the source features and the Frobenius norm is used as the matrix distance metric:

$$\min_{A} \|\boldsymbol{A}^T \boldsymbol{C_S} \boldsymbol{A} - \boldsymbol{C_T}\|^2 . \tag{6}$$

To avoid the expensive SVD steps, a regularization parameter λ is add to the diagonal elements of the covariance matrices $\boldsymbol{C_S}$ and $\boldsymbol{C_T}$ respectively. $\boldsymbol{I}$ denotes unit matrix with the same dimension of $\boldsymbol{C_S}$ and $\boldsymbol{C_T}$. The optimal $\boldsymbol{A}^*$ is given by [15]:

$$\boldsymbol{A}^* = (\boldsymbol{C_S} + \lambda_s \boldsymbol{I})^{-\frac{1}{2}} (\boldsymbol{C_T} + \lambda_t \boldsymbol{I})^{\frac{1}{2}}, \tag{7}$$

III. METHOD

This section describes the details of our mothed. The first stage, as shown in Figure 1, is used to extract features from raw EEG signals which include constructing multiple frequency-bands SCMs by filter banks, mapping the SCMs into Riemannian tangent space as a tangent vector and merging them in all band. The second stage, as in shown in Figure 2, aligns the calibration-session feature with the use-session feature by CORAL. Finally, using the aligned calibration-session feature to train a SVM classifier.

A. Filter Banks Riamenn Tangent Space Feature Extraction

We employ a filter bank that bandpass filters the *i-th* raw EEG trial $\boldsymbol{X_i} \in \mathbb{R}^{n \times T_s}$ into multiple bands $\boldsymbol{X_i}^{(1)}, \boldsymbol{X_i}^{(2)}, \dots, \boldsymbol{X_i}^{(8)}$ where $\boldsymbol{X_i}^{(j)} \in \mathbb{R}^{n \times T_s}$, j denotes *j-th* frequency band. The bandpass filter is 2-th order Butterworth filter. For each band, spatial covariance matrix $\boldsymbol{C_i}^{(j)}$ is computed by formula (1), Riemannian mean point $\boldsymbol{R}^{(j)}$ of spatial covariance matrices of all sample can be computed by [2]. Following the formula (4), at the tangent point $\boldsymbol{R}^{(j)}$, spatial covariance matrices can be mapped into their tangent space as tangent vectors $\boldsymbol{T_i}^{(j)}$. At the end, feature vector $\boldsymbol{T_i}$ is merged from tangent vectors $\boldsymbol{T_i}^{(j)}$ in all bands. In both calibration session and use session, this feature extraction procedure is same form. The filter banks Riemannian tangent space feature extraction is shown in Figure 1.

B. Align Feature Space

In subsection A, we obtain features of all calibration-session samples $\boldsymbol{F_S} = [\boldsymbol{T_1}, \boldsymbol{T_2}, \ldots, \boldsymbol{T_{Ns}}] \in \mathbb{R}^{d \times Ns}$ and features of all use-session samples $\boldsymbol{F_T} = [\boldsymbol{T_1}, \boldsymbol{T_2}, \ldots, \boldsymbol{T_{Nt}}] \in \mathbb{R}^{d \times Nt}$. Here, d denotes the dimension of features, Ns denotes the number of calibration-session samples and the Nt denotes the number of use-session samples. Following the formula (7), we can obtain the space transformation $\boldsymbol{A}^*$:

$$\boldsymbol{A}^* = (\boldsymbol{F_S} * \boldsymbol{F_S}^{\boldsymbol{T}} + \lambda_s \boldsymbol{I})^{-\frac{1}{2}} (\boldsymbol{F_T} * \boldsymbol{F_T}^{\boldsymbol{T}} + \lambda_t \boldsymbol{I})^{\frac{1}{2}}. \quad (8)$$

And we can map the features $\boldsymbol{F_S}$ into use-session feature space as [15], as is shown in Figure 2:

$$\boldsymbol{F_S}' = \boldsymbol{A}\boldsymbol{F_S} \quad (9)$$

Then we use the aligned calibration-session features $\boldsymbol{F_S}'$ with its labels to train a SVM classifier and test on use-session features $\boldsymbol{F_T}$.

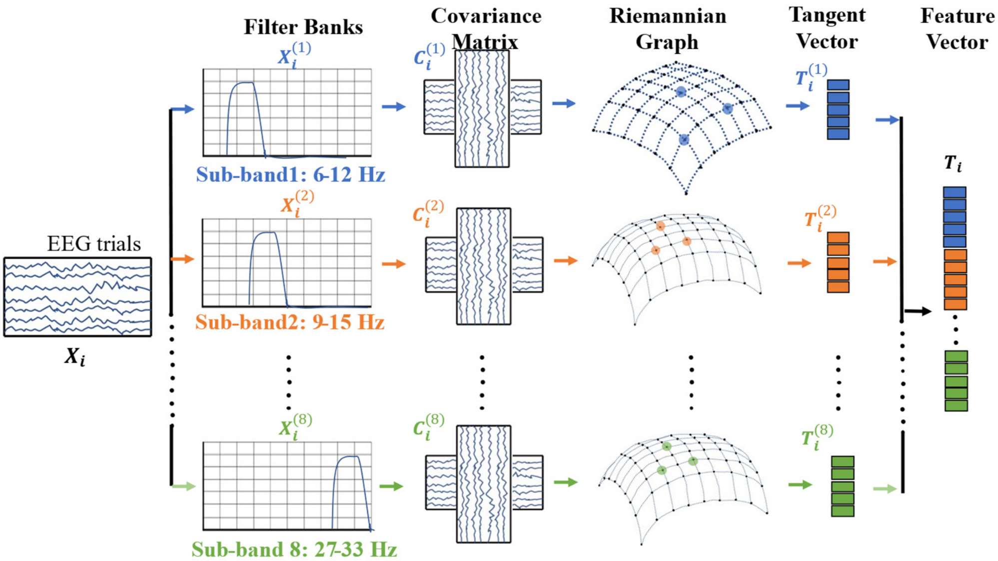

Figure 1. Mapping EEG signals into feature vector by Filter Banks Riemannian Tangent Space.

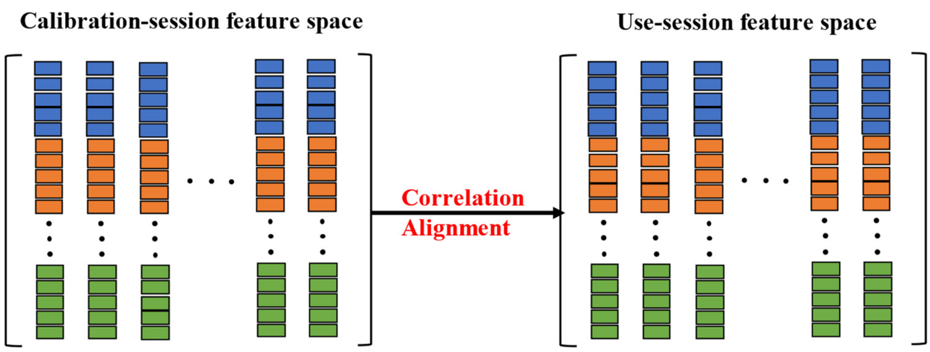

Figure 2. Aligning the calibration-session feature with the use-session feature by CORAL

IV. EXPERIMENT

A. Description of Data

To verify proposed method. We conduct experiment on 2008 BCI Competition IV 2a dataset (DS1), 2005 BCI Competition III 3a dataset (DS2) and 2005 BCI Competition III 4a dataset (DS3). The basic situation of the dataset is shown in Table 1. The detailed dataset description can be vied in [18] [21]. The Timing scheme of motor imagery is shown in Figure 3, the time interval between the two red dotted lines is short-time windows that we choose.

Table 1 The basic situation of the data set.

Dataset	DS1: BCI Competition IV 2a	DS2: BCI Competition III 3a	DS3: BCI Competition III 4a
Participant	9	3	5
category	4	4	3
Electrodes	22	same as DS1	same as DS1
Sampling Rate	250	250	100

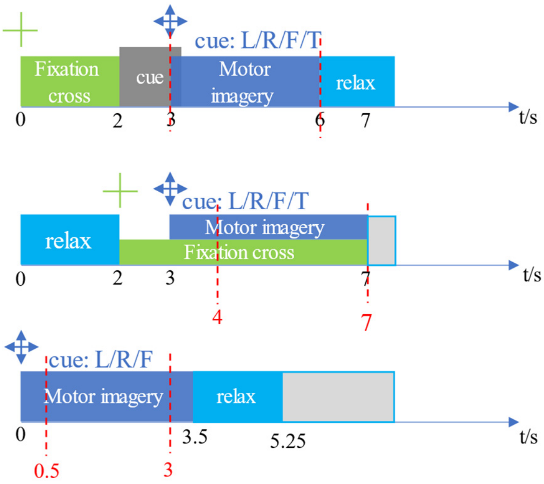

Figure 3. The Time scheme of the motor imagery paradigm for each trial on DS1, DS2 and DS3.

B. Paramerter Setting

The filter bank is set as 8 (such as, 6-12Hz,9-15Hz, …, 27-33Hz*)*. The filter is 2-th order Butterworth filter. All optimal hyperparameters are selected by cross-validation on the training set. The optimal parameters λ_s is set as 0.5, 0.5, and 0.004 respectively, the optimal parameters λ_t is 0.2, 0.0005, and 0.4 respectively, the optimal parameter C in SVM is 1.0, 0.6 and 2.1. The kernel used in SVM is a linear kernel. We directly use built-in SVM function in matlab2019b.

Table 2. Comparison of the performance of our methods on DS1, DS2 and DS3 for prediction on test data.

	Sub	Method			
		FBCSP	TSLDA [2]	TSSM [13]	FBTS CORAL
DS1	S1	0.68	0.74	0.77	**0.86**
	S2	0.42	0.38	0.33	**0.51**
	S3	0.75	0.72	0.77	**0.85**
	S4	0.48	0.5	0.51	**0.67**
	S5	0.4	0.26	0.35	**0.51**
	S6	0.27	0.34	0.36	**0.44**
	S7	0.77	0.69	0.71	**0.88**
	S8	0.75	0.71	0.72	**0.81**
	S9	0.61	0.76	**0.83**	0.69
	mean kappa	0.57	0.57	0.59	**0.69**
DS2	Sub	Method			
		1st	2st	3st	FBTS CORAL
	k3	0.82	0.90	0.94	**0.95**
	k6	**0.75**	0.43	0.41	**0.66**
	l1	**0.80**	0.71	0.52	**0.72**
	mean kappa	**0.79**	*0.68*	0.62	**0.78**
DS3	aa	**95.5**	89.3	82.1	79.5
	al	**100.0**	98.2	94.6	**100.0**
	av	**80.6**	76.5	70.4	72.4
	aw	**100.0**	92.4	87.5	**95.5**
	ay	**97.60**	80.6	88.1	**88.5**
	mean acc	**94.17**	85.12	83.45	**86.10**

C. Result

Table 2 reports the comparative classification performance on the DS1, DS2 and DS3. On DS1, the results from Correlation Alignment in Filter Bank Riemannian Tangent Space (**FBTSCORAL**) outperform existing the state-of-the-art performance. Proposed method has average 17.4% kappa improvements against **FBCSP** which won the first place in 2008 BCI competition IV 2a [18]. All p-values of paired T-tests on DS1 are smaller than 0.05, which proves proposed method have significantly high performance in small sample sizes. On DS2, we are only 1% behind the first place. On DS3, although we achieved the second place, the performance gap between proposed method and the state-of-the-art method is still relatively large. The possible reason is that the state-of-the-art method adopted complex ensemble model. Overall, proposed method effectively improves the classification performance of motor imagery in use-session.

Table 3 reports the ablation Study on DS1, DS2 and DS3 with accuracy. **FBTSCORAL** is proposed method. **FBTS** denotes that we extract multiple frequency bands features by Filter Bank Riemannian Tangent Space and directly identify motor intent by SVM. **TSCORAL** denotes that we extract a large frequency band (8-33 Hz) feature by Riemannian Tangent Space and align the calibration-session feature space with the use-session feature space by CORAL as well as SVM used to identify motor intent. The *TS* denotes that we extract a large frequency band (8-33 Hz) feature by Riemannian Tangent Space and directly identify motor intent by SVM. Comparing **TS** method and **TSCORAL** method, we can see that the unsupervised domain adaptation method CORA*L* indeed reduces the distribution difference to enhance classification performance. Comparing the **TS** and the **FBTS**, we can see that multiple frequency bands can capture discriminative information better than a large frequency band (8-33 Hz) to improve the model performance. Comparing the **TSCORAL** and the **FBTSCORAL**, we can see that conducting correlation alignment in multiple bands tangent space largely enhance classification performance for motor imagery, the **FBTSCORAL** achieves average 7.7%, 6.0% and 3.6% improvements against the **TSCORAL** on DS1, DS2 and DS3 respectively. Confusion matrix of DS3 is given in Figure 4. As expected, there are more misclassification samples between two hands and foot vs. tongue. Figure 4(a) is confusion matrix of **TS** method. Figure 4(b) is confusion matrix of the **TSCORAL** method. Figure 4(c) is confusion matrix of the **FBTS** method. Figure 4(d) is confusion matrix of the **FBTSCORAL** method. Comparing Figure 4(a) and Figure 4(d), we can see that proposed method effectively reduces confusion between two hands and foot vs. tongue.

Table 3. Ablation Study on DS1, DS2 and DS3 with accuracy (%).

	Sub	Method			
		TS	**TS CORAL**	**FBTS**	**FBTS CORAL**
DS1	**S1**	78.47	81.60	86.45	89.58
	S2	62.50	62.50	63.19	63.54
	S3	81.60	87.50	81.60	88.54
	S4	64.58	61.46	76.04	75.35
	S5	50.00	50.69	59.72	63.54
	S6	51.04	52.43	56.60	57.64
	S7	78.13	81.60	87.85	90.97
	S8	84.03	84.03	84.72	85.42
	S9	79.86	79.51	79.86	76.39
	mean acc	70.02	71.26	75.12	**76.77**
	std	13.3	14.4	12.1	12.7
DS2	**k3**	93.89	94.44	96.11	96.11
	k6	65.83	69.17	62.50	74.17
	l1	70.00	71.67	70.00	79.17
	mean acc	76.57	78.42	76.20	**83.15**
	std	15.1	13.9	17.6	11.5
DS3	**aa**	72.32	75.00	78.57	79.50
	al	100.0	100.00	100.00	100.00
	av	70.41	69.90	71.43	72.45
	aw	88.84	89.29	95.08	95.54
	ay	84.92	87.70	81.35	88.49
	mean acc	81.90	83.10	83.70	**86.07**
	std	12.2	12.0	12.0	11.2

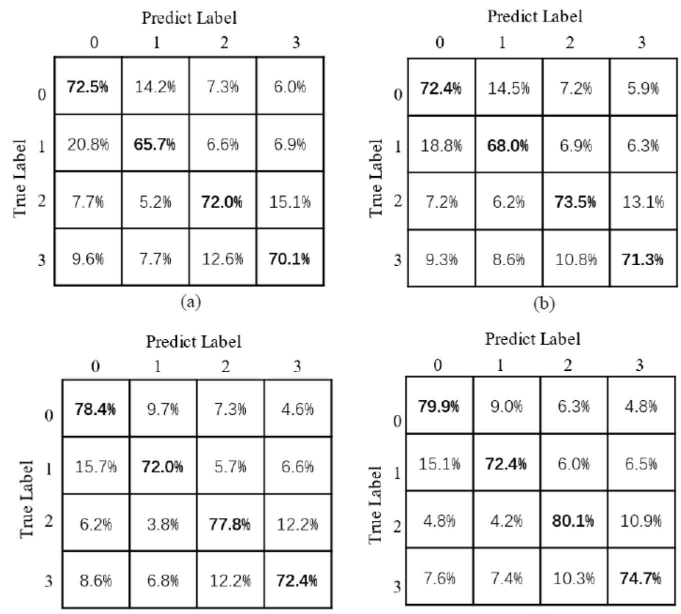

Figure 4. Confusion matrix in test dataset for average of DS3. (a). Confusion matrix of **TS** method. (b). Confusion matrix of **TSCORAL** method. (c). Confusion matrix of **FBTS** method. (d). Confusion matrix of **FBTSCORAL** method.

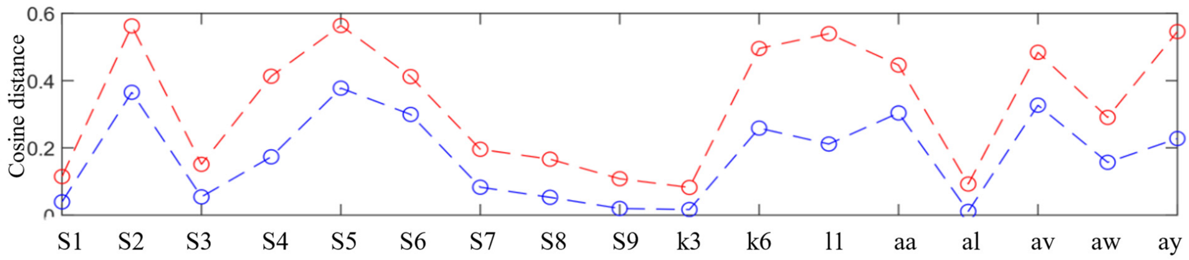

Figure 5. Distribution distance between source domain and target domain for each subject on DS1, DS2 and DS3. The red line denotes mean distance between source and target before alignment operation, the blue line denotes mean distance between source and target after alignment operation

In order to study whether CORAL aligns the source and target correctly in motor imagery, for each subject, we calculate mean distance of the source and target. As is shown in Figure 5, red line denotes mean distance between source domain and target domain before alignment operation, blue denotes mean distance between source and target after alignment operation, we can see CORAL can reduces the distribution distance between source domain and target domain in motor imagery.

V. Conclusion

We propose a Correlation Alignment in Filter Bank Riemannian Tangent Space method to align the calibration-session feature with the use-session feature in multiple bands Riemannian tangent space. Firstly, Filter Banks in Riemannian Tangent Space extracts multiple bands features in an unsupervised way which makes it very convenient for us to apply domain adaptation methods to reduce the distribution distance between source and target. Meanwhile, there are only a few parameters in our model, what we need set is just the regularization hyperparameter λ in Correlation Alignment and the kernel as well as the regularization hyperparameter of SVM. Our Model is efficient and stable. By combining unsupervised domain adaptive methods with Riemannian manifold, we enhance the motor imagery classification performance.

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